Algebra SEP Worksheet 9 8/15/23

Miscellaneous/Review

1. (August 2021 Problem 1) On this problem, only the answer will be graded.

(a) Put $R = \mathbb{Z}/12\mathbb{Z}$. For which elements $f \in R$ is the localization $R_f = 0$?

(b) Put $R = \mathbb{Z}/12\mathbb{Z}$. For which elements $f \in R$ is the localization R_f a field? (Recall that the zero ring is not a field.)

(c) The Jordan form of an operator $A : \mathbb{C}^6 \to \mathbb{C}^6$ is

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

What is the Jordan form of the operator $A^3 - A$?

1. Solution

(a) $f = 6$ since it is the only element such that $f^t = 0$ in R for some power t

(b) Recall that $R \to R_f$ is injective iff no power of f is a zero-divisor on R. In order for R_f to be a field, this map needs to not be injective. So, the zero-divisors on R (besides 6 since $R_6 = 0$) are our options. The zero-divisors are $2, 3, 4, 8, 9, 10$. Also, R has two prime ideals (2) and (3). In order for R_f to be a field, it needs to have exactly one prime ideal, so either (2) or (3) needs to become the unit ideal in R_f. But if (2) is the only surviving prime ideal, then $(4) \subsetneq (2)$ is another ideal in the localization, which would mean that it is not a field. So, we need to localize at an element in (2) so that (3) is the only surviving prime ideal, and we need (3) to become the zero ideal in the localization. Choosing $f = 2, 4, 8, 10$ works since then $\frac{3}{1} = 0$ in the localization.

(c)

$$A^3 - A = \begin{pmatrix} 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad \text{which has Jordan form:} \quad \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

2. (January 2022 Problem 1) On this problem, only the answer will be graded. Consider the symmetric group S_4.

(a) What is the order of S_4?

(b) What is the order of its center $Z(S_4)$?

(c) What is the order of its commutator subgroup $[S_4, S_4]$?

(d) Is S_4 a simple group?

(e) Is S_4 a solvable group?

(f) How many conjugacy classes does S_4 have?

2. Solution

(a) 24

(b) 1

(c) 12 ($[S_4, S_4] = A_4$; this is always the case)

(d) No, A_4 is a nontrivial normal subgroup.

(e) Yes, since A_4 contains a normal subgroup isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, and the quotient is isomorphic to $\mathbb{Z}/3\mathbb{Z}$.

(f) 5, one for each partition of 4.

3. (August 2022 Problem 1) On this problem, only your answer will be graded, no justification is necessary.

(a) Let $K = \mathbb{Q}(2^{1/3})$. Is $K/\mathbb{Q}$ a Galois extension?

(b) What is the radical of the ideal $(x^3 - x^2) \subset \mathbb{R}[x]$?

(c) How many different (up to isomorphism) groups of order 49 are there?

(d) Give an example of two non-zero abelian groups M, N such that $M \otimes_{\mathbb{Z}} N = 0$.

(e) How many two-sided ideals are there in $M_2(\mathbb{C})$, the ring of 2×2 matrices with entries in $\mathbb{C}$?

3. Solution

(a) No, since $\zeta_3 2^{1/3}$ is a root of $x^3 - 2$ which does not lie in K.

(b) $(x^2 - x)$, in this case radical corresponds to "largest squarefree divisor".

(c) 2, $\mathbb{Z}/49\mathbb{Z}$ and $\mathbb{Z}/7\mathbb{Z} \times \mathbb{Z}/7\mathbb{Z}$.

(d) $M = \mathbb{Z}/2\mathbb{Z}$, $N = \mathbb{Z}/3\mathbb{Z}$.

(e) 2, the zero ideal and the whole ring.

4. (August 2021 Problem 5) Let A be an algebra over $\mathbb{C}$ with two generators e and f and defining relations $e^2 = e, f^2 = f$.

(a) Let B be the ring of 2×2 matrices over the polynomial ring $\mathbb{C}[t]$. Consider B as an (infinite-dimensional) algebra over $\mathbb{C}$. Show that there exists a unique $\mathbb{C}$-algebra homomorphism $\phi : A \to B$ such that $\phi(e) = \begin{pmatrix} 1 & t \\ 0 & 0 \end{pmatrix}$ and $\phi(f) = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$.

(b) Prove that the homomorphism ϕ from part (a) is injective.

(c) Consider the algebra A as a vector space over $\mathbb{C}$. Write down a basis for A and prove that your answer is correct.

4. Solution

(a) We need to show that the map ϕ given is well-defined. So, we just need to show that it respects the relations in A, i.e. that $\phi(e^2 - e) = \phi(f^2 - f) = 0$. It is straightforword to compute that $\phi(e^2) = \phi(e)^2 = \phi(e)$ and similarly $\phi(f^2) = \phi(f)$, so this is a well-defined $\mathbb{C}$-algebra homomorphism $A \to B$. It is unique because it is defined on the generators of A.

(b) Any element in A can be written as a $\mathbb{C}$-linear combination of $\{1, e, f, ef, fe, efe, fef, ...\}$. So, let's think about the image of these elements under ϕ. We find that $\phi(efe) = t\phi(e)$, $\phi(fef) = t\phi(f)$, $\phi(efef) = t^2\phi(ef)$, $\phi(fefe) = t^2\phi(fe)$,..., $\phi(efef \cdots e) = t^n\phi(e)$, $\phi(fefe \cdots f) = t^n\phi(f)$, $\phi(efef \cdots f) = t^n\phi(ef)$, and $\phi(fefe \cdots e) = t^n\phi(fe)$ for some n. Therefore, $\phi(a)$ for any $a \in A$ can be written as

$$\phi(a) = \begin{pmatrix} a_1 & 0 \\ 0 & a_1 \end{pmatrix} + p_e(t)\begin{pmatrix} 1 & t \\ 0 & 0 \end{pmatrix} + p_f(t)\begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} + p_{ef}(t)\begin{pmatrix} t & t \\ 0 & 0 \end{pmatrix} + p_{fe}(t)\begin{pmatrix} 0 & 0 \\ 1 & t \end{pmatrix}$$

where the $p_i(t) \in \mathbb{C}[t]$ are determined by the coefficients of the elements in a. Therefore, if $\phi(a) = 0$, we have

$$\begin{pmatrix} a_1 + p_e(t) + tp_{ef}(t) & t(p_e(t) + p_{ef}(t)) \\ p_f(t) + p_{fe}(t) & a_1 + p_f(t) + tp_{fe}(t) \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Solving this system gives that $p_e(t) = p_f(t) = -p_{ef}(t) = -p_{fe}(t)$, which then gives that $a_1 + (1 - t)p_e(t) = 0$. Considering the degrees of the polynomials, we see this is only possible if $p_e(t) = 0$, and then $a_1 = 0$. Thus, $a = 0$, and ϕ is injective.

(c) A basis for A as a $\mathbb{C}$-vector space is given by $\{1, e, f, ef, fe, efe, fef, \cdots\}$, so this is an infinite-dimensional $\mathbb{C}$-vector space.

5. (August 2016 Problem 4) Let R be the subring of $\mathbb{C}[x, y]$ consisting of the polynomials $P(x, y)$ such that $P(x, y) = P(y, x)$.

(a) Show that R is generated as a $\mathbb{C}$-algebra by $x + y$ and xy.

(b) Show that the map $\mathbb{C}[u, v] \to R$ sending u to $x + y$ and v to xy is an isomorphism.

(c) Let S be the subring of $\mathbb{C}[x, y]$ consisting of the polynomials $P(x, y)$ such that $P(x, y) = P(-x, -y)$. Can S be generated as a $\mathbb{C}$-algebra by two polynomials? Either give two generators, or prove that S is not generated by 2 polynomials.

5. Solution

(a) Let $P(x, y) \in R$. By induction on the degree of P, we may assume that P is homogeneous of some degree n and that all lower degree polynomials in R are generated by $x + y$ and xy over $\mathbb{C}$. Because $P(x, y) = P(y, x)$, we must have that the number of x factors is the same as the number of y factors. So, if x^t divides P for some $t > 0$, then y^t divides P, but then $(xy)^t$ divides P, and we are done by the induction hypothesis. Therefore, we can assume that $t = 0$, i.e. no power of x or y divides P. In this case, we must have $P = x^n + y^n + xyQ(x, y)$ for some symmetric $Q(x, y)$ of lesser degree. But then $P - (x + y)^n$ is divisible by xy.

(b) Part (a) gives that this map is surjective. So we just need to show injectivity. Suppose $Q(u, v) \in \mathbb{C}[u, v]$ is in the kernel, i.e. $Q(x + y, xy) = 0$, and assume that $Q(u, v) \neq 0$. Observe that taking $y = 0$ gives $Q(x, 0) = 0$, which implies $Q(u, 0) = 0$, so every term of Q has a factor of v. Similarly, taking $x = 0$ gives that $Q(y, 0) = 0$ and by symmetry in R, we have $Q(0, y) = 0$, so every term of Q has a factor of u as well. Therefore, by unique factorization, we have $Q(u, v) = uvQ'(u, v)$. Applying the map gives $(x+y)(xy)Q'(x+y, xy) = 0$, which implies that $Q'(x+y, xy) = 0$ since $(x+y)xy \neq 0$ and R is a domain. This means that $Q'(u, v)$ is an element of the kernel of strictly smaller degree than $Q(u, v)$. But then we could just repeat the argument with $Q'(u, v)$ to produce infinitely many elements of the kernel of smaller and smaller degree, which is impossible. Thus, $Q(u, v) = 0$.

(c) We first show that $S = \mathbb{C}[x^2, xy, y^2]$. The backwards containment is clear. For the forward containment, note that if $P(x, y) \in S$, then $P(x, x) = P(-x, -x)$ is a polynomial is x where all terms must have even degree. Therefore, every term of $P(x, y)$ must have even total degree. So every monomial in $P(x, y)$ is of the form $x^i y^j$ with $i + j$ even and WLOG $i \geq j$. Then $x^i y^j = (xy)^j x^{i-j} \in \mathbb{C}[x^2, xy, y^2]$ since $i - j$ is also even. Then $S = \mathbb{C}[x^2, xy, y^2]$ cannot be generated as a $\mathbb{C}$-algebra by two polynomials because it is 3-dimensional as an algebra over $\mathbb{C}$.

6. (August 2014 Problem 4) Let R be a commutative ring and M an R-module. Recall that a prime ideal P is an associated prime of M if there exists some $m \in M$ such that P consists of those f such that $fm = 0$; i.e. P is the annihilator of some $m \in M$.

(a) Let $0 \to M' \to M \to M'' \to 0$ be an exact sequence of R-modules, and let P be an associated prime of M. Prove that P is an associated prime of either M' or M''.

(b) Is the converse true? That is, if P is an associated prime of either M' or M'', must it be the case that P is an associated prime of M?

6. Solution

(a) Suppose P is not an associated prime of M', and write $P = \mathrm{ann}(m)$ for some $m \in M$. We want to show that $P = \mathrm{ann}(m'')$ for some $m'' \in M''$. Let ϕ be the map $M \to M''$ in the SES. Then a natural choice is $m'' = \phi(m)$. We will show that this does indeed work. Let $f \in P$ so that $fm = 0$. Then $f\phi(m) = \phi(fm) = 0$, so $f \in \mathrm{ann}(\phi(m))$. Thus, $P \subseteq \mathrm{ann}(\phi(m))$. Conversely, if $f \in \mathrm{ann}(\phi(m))$, then $\phi(fm) = f\phi(m) = 0$, so $fm \in \ker(\phi) = \mathrm{im}(\varphi)$ by the exactness of the sequence where $\varphi : M' \to M$. Suppose towards a contradiction that $f \notin P$. Then $fm = \varphi(m') \neq 0$, so $m' \neq 0$ since φ is injective. We will show that this gives $P = \mathrm{ann}(m')$, contradicting our assumption. For any $g \in P$, we have $\varphi(gm') = g\varphi(m') = g(fm) = f(gm) = 0$, so $gm' = 0$ by injectivity, which gives $g \in \mathrm{ann}(m')$. On the other hand, given $h \in \mathrm{ann}(m')$, we have $0 = \varphi(hm') = h\varphi(m') = hfm$, so $hf \in P = \mathrm{ann}(m)$. Since $f \notin P$ and P is prime, we must have $h \in P$. This gives our contradiction; thus, $f \in P$ and $P = \mathrm{ann}(\phi(m))$.

(b) No, take for example the SES of $\mathbb{Z}$-modules $0 \to \mathbb{Z} \to \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$ where the first map is multiplication by 2. Then (2) is the annihilator of 1 in $\mathbb{Z}/2\mathbb{Z}$, but $\mathbb{Z}$ is torsion-free, so its only associated prime is (0).

7. (August 2019 Problem 3) Let L/K be a Galois extension of fields with Galois group G. Let S be the subset of roots of unity in L; that is, S is the set of elements $x \in L$ such that $x^k = 1$ for some positive integer k.

(a) Show that the action of G on L restricts to an action of G on S (i.e., the action of G preserves S.)

(b) Let $L = \mathbb{Q}(i)$ and $K = \mathbb{Q}$. Show in this case that G acts faithfully on S.

(c) Give an example of L/K such that the action of G on S is not faithful.

(d) Suppose that the action of G on S is transitive. Prove that $\mathrm{char}(K) = 2$.

7. Solution

(a) The nth roots of unity in L are precisely the elements in L that are roots of the equations $t^n - 1$. But $t^n - 1 \in K[t]$, so G must permute the roots of this polynomial. Thus, G also acts on S.

(b) In this case, $S = \{\pm 1, \pm i\}$ and since $[L : K] = 2$, $G \cong \mathbb{Z}/2\mathbb{Z}$. So the action of G on S is given by 0 acting trivially and 1 acting by $i \mapsto -i$. So this action is faithful since only the identity element in G acts trivially on S.

(c) Let $L = \mathbb{Q}(\sqrt{2})$ and $K = \mathbb{Q}$. Then $S = \{\pm 1\}$ since those are the only roots of unity in $\mathbb{R}$ and $L \subseteq \mathbb{R}$. As in the previous example G is a cyclic group of order 2. But both elements of G fix S since $S \subseteq \mathbb{Q}$.

(d) Imagine K has characteristic not equal to 2. Then -1 and 1 are distinct elements of $S \cap K$. But every element of G will fix both -1 and 1 since they are in K, so the action could not be transitive.